

MATH 200 - Elementary Linear Algebra

MATH 201 - Intermediate Linear Algebra

Section 6.3 Notes

Matrices for Linear Transformation

Devotional Thought

Standard Matrix for a Linear Transformation

In section 6.1 we saw that matrix multiplication of a column vector by a matrix (on the left) is always a linear transformation (Theorem 6.2).

Which representation of $T : R^3 \rightarrow R^3$ is better:

$$T(x_1, x_2, x_3) = (2x_1 + x_2 - x_3, -x_1 + 3x_2 - 2x_3, 3x_2 + 4x_3)$$

or

$$T(\mathbf{x}) = A\mathbf{x} = \begin{bmatrix} 2 & 1 & -1 \\ -1 & 3 & -2 \\ 0 & 3 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} ?$$

The second representation is better than the first for at least three reasons: it is simpler to write, simpler to read, and easier to enter into a calculator or math software. Later on we will also see that matrix representation of a linear transformation has some theoretical advantages.

In this section, we will see that for linear transformations involving finite-dimensional vector spaces, matrix representation is always possible.

The key to representing a linear transformation $T : V \rightarrow W$ by a matrix is to determine how it acts on a basis for V . Once you know the image of every vector in the basis, you can use the properties of linear transformations to determine $T(\mathbf{v})$ for any $\mathbf{v}$ in V .

Recall The standard basis for R^n , written in column vector notation, is

$$B = \{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\} = \left\{ \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}, \dots, \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix} \right\}.$$

Theorem 6.10 Standard Matrix for a Linear Transformation

Let $T : R^n \rightarrow R^m$ be a linear transformation such that, for the standard basis vectors $\mathbf{e}_i$ of R^n ,

$$T(\mathbf{e}_1) = \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix}, T(\mathbf{e}_2) = \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix}, \dots, T(\mathbf{e}_n) = \begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix}.$$

Then the $m \times n$ matrix whose n columns correspond to $T(\mathbf{e}_i)$

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

is such that $T(\mathbf{v}) = A\mathbf{v}$ for every $\mathbf{v}$ in R^n . A is called the **standard matrix** for T .

Proof of Theorem 6.10. Let $\mathbf{v}$ be an arbitrary vector in R^n . Then

$$\mathbf{v} = \begin{bmatrix} v_1 & v_2 & \cdots & v_n \end{bmatrix} = v_1\mathbf{e}_1 + v_2\mathbf{e}_2 + \cdots + v_n\mathbf{e}_n.$$

Thus, since T is a linear transformation,

$$\begin{aligned} T(\mathbf{v}) &= T(v_1\mathbf{e}_1 + v_2\mathbf{e}_2 + \cdots + v_n\mathbf{e}_n) \\ &= v_1T(\mathbf{e}_1) + v_2T(\mathbf{e}_2) + \cdots + v_nT(\mathbf{e}_n) \\ &= v_1 \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix} + v_2 \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix} + \cdots + v_n \begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix} \\ &= \begin{bmatrix} a_{11}v_1 + a_{12}v_2 + \cdots + a_{1n}v_n \\ a_{21}v_1 + a_{22}v_2 + \cdots + a_{2n}v_n \\ \vdots \\ a_{m1}v_1 + a_{m2}v_2 + \cdots + a_{mn}v_n \end{bmatrix} \\ &= \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} \\ &= A\mathbf{v}. \end{aligned}$$

Therefore $T(\mathbf{v}) = A\mathbf{v}$ for all $\mathbf{v}$ in R^n . ■

That is, the converse of Theorem 6.2 holds: A function $T : R^n \rightarrow R^m$ is a linear transformation if and only if T is given by $T(\mathbf{v}) = A\mathbf{v}$ for some $m \times n$ matrix A .

Example [Example 1] Find the standard matrix for the linear transformation $T : R^3 \rightarrow R^2$ defined by $T(x, y, z) = (x - 2y, 2x + y)$.

Question What is the standard matrix for the zero transformation from R^n into R^m ?

What is the standard matrix for the identity transformation from R^n into R^n ?

Example [Exercise 12] Let T be the reflection in the line $y = x$ in R^2 :

$$T(x, y) = (y, x)$$

(a) Find the standard matrix for the linear transformation T .

(b) Use A to find the image of the vector $\mathbf{v} = (3, 4)$.

Compositions of Linear Transformations and their Matrices

Recall that we can *compose* functions f and g as $(f \circ g)(x) = f(g(x))$.
(e.g. Chain Rule/ u -substitution)

Function composition is known for being noncommutative: in general, $f \circ g \neq g \circ f$.

But we also know that AB need not equal BA for matrices A and B .

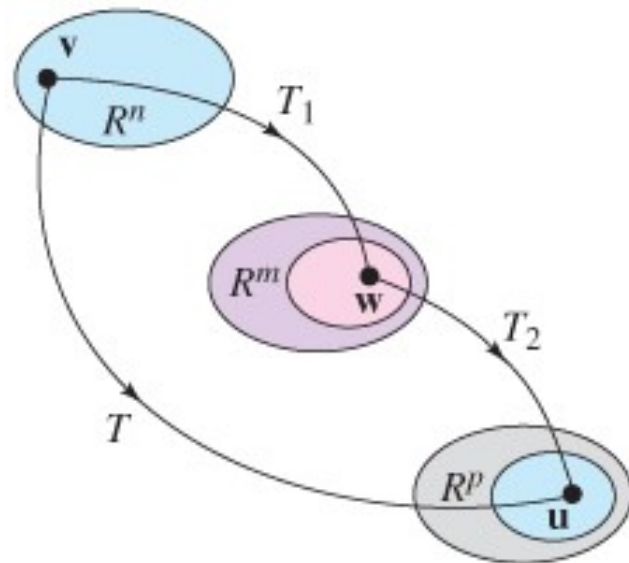
The **composition**, T , of $T_1 : R^n \rightarrow R^m$ with $T_2 : R^m \rightarrow R^p$ is

$$T(\mathbf{v}) = T_2(T_1(\mathbf{v})).$$

where $\mathbf{v}$ is a vector in R^n . This composition is denoted by

$$T = T_2 \circ T_1.$$

The domain of T is the domain of T_1 . Moreover, the composition is not defined unless the range of T_1 lies within the domain of T_2 .



The next theorem not only states that the composition of two linear transformations is a linear transformation, but also says that the standard matrix for the composition is the product of the standard matrices for the two original linear transformations.

Theorem 6.11 Composition of Linear Transformations Let $T_1 : R^n \rightarrow R^m$ and $T_2 : R^m \rightarrow R^p$ be linear transformations with standard matrices A_1 and A_2 , respectively. The composition $T : R^n \rightarrow R^p$, defined by $T(\mathbf{v}) = T_2(T_1(\mathbf{v}))$, is a linear transformation. Moreover, the standard matrix A for T is the matrix product $A = A_2A_1$.

Proof. Let $\mathbf{u}$ and $\mathbf{v}$ be vectors in R^n . Thus, since T_1 and T_2 are linear transformations,

$$\begin{aligned} T(\mathbf{u} + \mathbf{v}) &= T_2(T_1(\mathbf{u} + \mathbf{v})) \\ &= T_2(T_1(\mathbf{u}) + T_1(\mathbf{v})) \\ &= T_2(T_1(\mathbf{u})) + T_2(T_1(\mathbf{v})) \\ &= T(\mathbf{u}) + T(\mathbf{v}). \end{aligned}$$

Similarly, for an arbitrary vector $\mathbf{u}$ in R^n and an arbitrary scalar c ,

$$\begin{aligned} T(c\mathbf{u}) &= T_2(T_1(c\mathbf{u})) \\ &= T_2(cT_1(\mathbf{u})) \\ &= cT_2(T_1(\mathbf{u})) \\ &= cT(\mathbf{u}). \end{aligned}$$

So T is a linear transformation.

Furthermore, for every $\mathbf{v}$ in R^n ,

$$T(\mathbf{v}) = T_2(T_1(\mathbf{v})) = T_2(A_1\mathbf{v}) = A_2(A_1\mathbf{v}) = (A_2A_1)\mathbf{v}.$$

So A_2A_1 is the standard matrix for T . ■

Example [Example 3] Let T_1 and T_2 be linear transformations from R^3 into R^3 such that

$$T_1(x, y, z) = (2x + y, 0, x + z) \quad \text{and} \quad T_2(x, y, z) = (x - y, z, y).$$

Find the standard matrices for the compositions $T = T_2 \circ T_1$ and $T' = T_1 \circ T_2$.

The Inverse of a Linear Transformations

Another benefit of matrix representation is that it can represent the *inverse* of a linear transformation.

Definition If $T_1 : R^n \rightarrow R^n$ and $T_2 : R^n \rightarrow R^n$ are linear transformations such that for every $\mathbf{v}$ in R^n

$$T_2(T_1(\mathbf{v})) = \mathbf{v} \quad \text{and} \quad T_1(T_2(\mathbf{v})) = \mathbf{v},$$

then T_2 is the **inverse** of T_1 , and T_1 is said to be **invertible**.

Not every linear transformation has an inverse. If the transformation T_1 is invertible, however, then the inverse is *unique* and is denoted by T_1^{-1} .

Just as the inverse of a function of a real variable can be thought of as undoing what the function did, the inverse of a linear transformation T can be thought of as undoing the mapping done by T . For example, if T is a linear transformation from R^3 into R^3 such that

$$T(1, 4, -5) = (2, 3, 1)$$

and if T^{-1} exists, then T^{-1} maps $(2, 3, 1)$ back to its preimage under T . That is,

$$T^{-1}(2, 3, 1) = (1, 4, -5).$$

Theorem 6.12 Existence of an Inverse Transformation

Let $T : R^n \rightarrow R^n$ be a linear transformation with standard matrix A . Then T is invertible if and only if A is invertible. Additionally, if T is invertible with standard matrix A , then the standard matrix for T^{-1} is A^{-1} .

Example [Example 4] Consider the linear transformation $T : R^3 \rightarrow R^3$ defined by

$$T(x_1, x_2, x_3) = (2x_1 + 3x_2 + x_3, 3x_1 + 3x_2 + x_3, 2x_1 + 4x_2 + x_3).$$

Show that T is invertible, and find its inverse.

Example [Exercise 9] Determine whether the linear transformation is invertible. If it is, find its inverse.

$$T(x, y) = (x + y, 3x + 3y)$$

Transformation Matrix for a Nonstandard Basis

We now consider the more general problem of finding a matrix A for a linear transformation $T : V \rightarrow W$, where B and B' are ordered bases for V and W , respectively. Recall that the coordinate matrix of $\mathbf{v}$ relative to B is denoted by $[\mathbf{v}]_B$. To represent the linear transformation T , multiply A by a *coordinate matrix relative to B* to obtain a *coordinate matrix relative to B'* . That is, $[T(\mathbf{v})]_{B'} = A [\mathbf{v}]_B$. The matrix A is called the **matrix of T relative to the bases B and B'** .

Transformation Matrix for Nonstandard Bases

Let V and W be finite-dimensional vector spaces with bases B and B' , respectively, where

$$B = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}.$$

If $T : V \rightarrow W$ is a linear transformation such that

$$[T(\mathbf{v}_1)]_{B'} = \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix}, [T(\mathbf{v}_2)]_{B'} = \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix}, \dots, [T(\mathbf{v}_n)]_{B'} = \begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix},$$

then the $m \times n$ matrix whose n columns correspond to $T(\mathbf{v}_i)$

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

is such that $[T(\mathbf{v})]_{B'} = A [\mathbf{v}]_B$ for every $\mathbf{v}$ in V .

Example [Examples 5 and 6] Let $T : R^2 \rightarrow R^2$ be a linear transformation defined by $T(x_1, x_2) = (x_1 + x_2, 2x_1 - x_2)$. Find the matrix for T relative to the bases

$$B = \{(1, 2), (-1, 1)\} \quad \text{and} \quad B' = \{(1, 0), (0, 1)\}.$$

Use the matrix A to find $T(\mathbf{v})$, where $\mathbf{v} = (2, 1)$.